

Chapter 2: Vector spaces

Vector spaces provide an abstraction of geometric ideas like “points”, “lines” and “planes”. They capture the structure of the set of solutions of a homogeneous system of linear equations. Vector spaces appear naturally in all of mathematics, and feature in many applications.

The abstract notion allows to deal with all examples *at the same time*, rather than having to understand each individual case separately.

We will mostly concentrate on the abstract side and how it connects to calculations with systems of linear equations. The geometry will be used as motivating example.

Section 2.1: Idea and definition

1 Arrows in the plane

We want to consider the collection V of arrows in the plane, disregarding the starting point of the arrow. That is, two arrows are considered equal if they have the same length and the same direction:

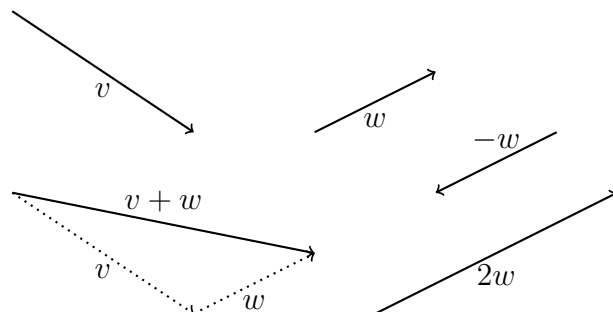


These five arrows represent three distinct elements of V (the first, third and fourth arrow have all the same length and direction, so represent the same element).

There are two natural operations with arrows:

- Add two arrows: form $v + w$ by concatenation, *i.e.*, place the start of w on the end of v , and let the sum be the arrow going from the start of v to what is now the end of w .
- Scaling an arrow: form λv , for $\lambda \in \mathbb{R}$, by stretching the arrow v by a factor λ ; if $\lambda < 0$ this also means that the arrow reverses its orientation.

2 Arrows in the plane (continued 1)



There is the “zero arrow”, written $\underline{0}$, which has length 0 and is hard to draw as an arrow (as start and end are the same).

We can think of an arrow as a pair $\begin{pmatrix} a \\ b \end{pmatrix}$ of real numbers; this means “going distance a in x -direction, and distance b in y -direction”.

3 Arrows in the plane (continued 2)

These operations on V satisfy the following properties — this is elementary geometry, and we won’t go through it in detail:

-
- **(A1)** $u + v = v + u$ for all $u, v \in V$;
 - **(A2)** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$;
 - **(A3)** $v + \underline{0} = v$ for all $v \in V$;
 - **(A4)** for each $v \in V$ there exists an element $w \in V$ such that $v + w = \underline{0}$;
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- **(S1)** $\lambda(u + v) = \lambda u + \lambda v$ for all $u, v \in V$ and all $\lambda \in \mathbb{R}$;
- **(S2)** $(\lambda + \mu)v = \lambda v + \mu v$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;
- **(S3)** $(\lambda\mu)v = \lambda(\mu v)$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;

- (S4) $1v = v$, for all $v \in V$.

In (A4), the arrow w is just v reversed, that is, $w = (-1)v$.

4 Solutions of homogeneous systems

Consider the following homogeneous system of linear equations:

$$\left. \begin{array}{rrrr} 2x & + & 3y & - & z & = & 0 \\ 3x & + & 3y & + & 2z & = & 0 \end{array} \right\} \quad (\text{H})$$

Let $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ be two solutions of (H), and let $\lambda \in \mathbb{R}$. Then:

- $\underline{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ is a solution of (H).
- $\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a solution of (H), where $\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x+a \\ y+b \\ z+c \end{pmatrix}$.
- $\lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is a solution of (H), where $\lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \\ \lambda z \end{pmatrix}$.

For example, $x+a$, $y+b$ and $z+c$ satisfy the first equation of the system (H):

$$2(x+a) + 3(y+b) - (z+c) = \underbrace{(2x + 3y - z)}_{=0} + \underbrace{(2a + 3b - c)}_{=0} = 0$$

5 Solutions of homogeneous systems (continued)

Thus we can “add” two solutions, and multiply a solution by a real number, and obtain another solution.

These operations on the set V of all solutions of (H) satisfy the following properties (as one can verify easily):

- (A1) $u + v = v + u$ for all $u, v \in V$;
 - (A2) $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$;
 - (A3) $v + \underline{0} = v$ for all $v \in V$;
 - (A4) for each $v \in V$ there exists an element $w \in V$ such that $v + w = \underline{0}$;
-

- (S1) $\lambda(u + v) = \lambda u + \lambda v$ for all $u, v \in V$ and all $\lambda \in \mathbb{R}$;
- (S2) $(\lambda + \mu)v = \lambda v + \mu v$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;
- (S3) $(\lambda\mu)v = \lambda(\mu v)$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;
- (S4) $1v = v$, for all $v \in V$.

6 Vector spaces: definition

A vector space is an object that behaves just like the examples of “arrows in the plane” and “solutions of a system of linear equations”: we can add two elements of the vector space, for example, and the conditions (A1)–(S4) are satisfied.

Definition: A *vector space over $\mathbb{R}$* , or *$\mathbb{R}$ -vector space*, is a set V together with

- a distinguished element $\underline{0} \in V$,
- an *addition* operation: given $u, v \in V$ we get $u + v \in V$, and
- a *multiplication by a scalar* operation: given $v \in V$ and $\lambda \in \mathbb{R}$ we have $\lambda v \in V$,

such that certain conditions (A1)–(A4) and (S1)–(S4) hold.

7 Vector spaces: definition (continued)

These conditions are the following:

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- **(A1)** $u + v = v + u$ for all $u, v \in V$;
 - **(A2)** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$;
 - **(A3)** $v + \underline{0} = v$ for all $v \in V$;
 - **(A4)** for each $v \in V$ there exists an element $w \in V$ such that $v + w = \underline{0}$;
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- **(S1)** $\lambda(u + v) = \lambda u + \lambda v$ for all $u, v \in V$ and all $\lambda \in \mathbb{R}$;
- **(S2)** $(\lambda + \mu)v = \lambda v + \mu v$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;
- **(S3)** $(\lambda\mu)v = \lambda(\mu v)$ for all $v \in V$ and all $\lambda, \mu \in \mathbb{R}$;
- **(S4)** $1v = v$, for all $v \in V$.

8 Vector spaces: terminology

Let V be an $\mathbb{R}$ -vector space as defined above.

- The elements of V are called *vectors*.
- The distinguished element $\underline{0} \in V$ is called the *zero vector*.
- We may say “*linear space over $\mathbb{R}$* ” instead of “vector space over $\mathbb{R}$ ”.
- When talking about an $\mathbb{R}$ -vector space, real numbers are often called *scalars*.

A *vector* is an element of a vector space.

(This is the *definition* of what a vector is.)

Section 2.2: Examples of vector spaces

9 The very first examples

1. The set of vectors in the plane (slides 1–3 of this chapter), with addition by concatenation, multiplication by a scalar given by dilation, and zero arrow $\underline{o}$ having start and end point the same, is a vector space.
2. In a similar manner, one can consider arrows in 3-space.
3. The set of real numbers $V = \mathbb{R}$, equipped with the usual addition and multiplication, and with $\underline{o} = 0$, is a vector space over $\mathbb{R}$. (Geometrically, this is “arrows in the number line.”)

10 The trivial vector space

A vector space must have at least one element, namely the zero vector.

But this is all it needs: The one-element set $\{\underline{o}\}$ is an $\mathbb{R}$ -vector space if we *define* operations as follows:

$$\begin{aligned}\underline{o} + \underline{o} &= \underline{o} \\ \lambda \underline{o} &= \underline{o} \quad \text{for all } \lambda \in \mathbb{R}\end{aligned}$$

The zero vector of $\{\underline{o}\}$ is, of course, $\underline{o}$.

This structure satisfies conditions (A1)–(S4) (easy to check).

This vector space is called the *trivial vector space*.

11 Ordered n -tuples

Let $n \in \mathbb{N}$. An *ordered n -tuple of real numbers* is a collection $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

where $x_i \in \mathbb{R}$ for each $i = 1, \dots, n$.

The n -tuples $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ are equal if corresponding entries are all equal: $x_1 = y_1$ and $x_2 = y_2$ and $\dots$ and $x_n = y_n$.

Ordered 2-tuples ($n = 2$) are just ordered pairs of real numbers, ordered 1-tuples are (real) numbers.

12 The vector space $\mathbb{R}^n$

$\mathbb{R}^n$ is the set of ordered n -tuples of real numbers:

$$\mathbb{R}^n = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n \right\}.$$

Addition and multiplication by scalar:

$$\text{For } x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n, \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n \quad \text{and} \quad \lambda \in \mathbb{R},$$

we define

$$x + y = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix} \quad \text{and} \quad \lambda x = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \end{pmatrix}.$$

13 The vector space $\mathbb{R}^n$ (continued)

Define $\underline{0}$ to be the “all zero” n -tuple:

$$\underline{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

The set $\mathbb{R}^n$ equipped with this zero vector and these operations is a vector space over $\mathbb{R}$.

Proof. This is quite obvious: $\mathbb{R}$ is a vector space, and the operations in $\mathbb{R}^n$ are defined componentwise. So the conditions (A1)–(S4) hold for $\mathbb{R}^n$ because they hold for $\mathbb{R}$. ■

14 Homogeneous systems of linear equations

Recall that a system of linear equations is called *homogeneous* if the right-hand side of every equation of the system is 0:

$$\left. \begin{array}{ccccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & 0 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & 0 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & 0 \end{array} \right\} \quad (\text{H})$$

Define the set of solutions of the system (H) by

$$L = \left\{ x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n \mid x_1, x_2, \dots, x_n \text{ are a solution of (H)} \right\}.$$

Note that $L \subseteq \mathbb{R}^n$.

15 Solutions of a homogeneous system

Recall: $\underline{0} \in \mathbb{R}^n$ is the “all-zero n -tuple”.

Claim: For the homogeneous system (H) we have $\underline{0} \in L$.

Proof. We have $a_{i,1}0 + a_{i,2}0 + \dots + a_{i,n}0 = 0$ for every i , so that $x_1 = x_2 = \dots = x_n = 0$ is a solution of every equation in H . ■

A homogeneous system of linear equations always has **at least one** solution, namely the trivial solution (that is, all variables zero).

16 Solutions of (H) form a vector space

Let L be the set of solutions of the homogeneous system (H) as above.

The set L , equipped with zero vector, addition and multiplication by scalar inherited from $\mathbb{R}^n$, is a vector space.

Proof. This is *almost* trivial as all the properties (A1) – (S4) for L follow from the fact that $\mathbb{R}^n$ satisfies these properties. The only thing to check is that $\underline{0} \in L$, which we did already, and that for all $\lambda \in \mathbb{R}$ and $x, y \in L$ we have $\lambda x \in L$ and $x + y \in L$.

17 Solutions of (H) form a vector space (continued)

$$\lambda \in \mathbb{R}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in L \quad \text{and} \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in L$$

For all $i = 1, 2, \dots, m$ we have:

$$\begin{aligned} a_{i,1}x_1 + a_{i,2}x_2 + \dots + a_{i,n}x_n &= 0 \\ a_{i,1}y_1 + a_{i,2}y_2 + \dots + a_{i,n}y_n &= 0 \end{aligned} \quad (*)$$

By adding we get

$$a_{i,1}(x_1 + y_1) + a_{i,2}(x_2 + y_2) + \dots + a_{i,n}(x_n + y_n) = 0$$

which means $x + y \in L$, and by multiplying (*) by λ we get

$$a_{i,1}(\lambda x_1) + a_{i,2}(\lambda x_2) + \dots + a_{i,n}(\lambda x_n) = 0$$

which means $\lambda x \in L$. ■

18 Sequences of real numbers

$$\mathcal{S} = \{(a_n)_{n=1}^{\infty} \mid \text{all } a_n \in \mathbb{R}\} = \text{set of sequences of real numbers}$$

For $a = (a_n)_{n=1}^{\infty}$, $b = (b_n)_{n=1}^{\infty}$ and $\lambda \in \mathbb{R}$ we define

- the sum sequence $a + b = (a_n + b_n)_{n=1}^{\infty}$,
- the scalar multiple sequence $\lambda a = (\lambda a_n)_{n=1}^{\infty}$,

and denote by $\underline{0}$ the “all zero” sequence $(0)_{n=1}^{\infty} \in \mathcal{S}$.

The set $\mathcal{S}$ equipped with this zero vector $\underline{0}$ and these operations is a vector space over $\mathbb{R}$.

Informal proof: a sequence is an “ordered tuple” with infinitely many entries. In this sense, $\mathcal{S}$ is an “infinite” version of $\mathbb{R}^n$. The result then follows as operations are defined “entrywise”.

19 Matrices

Let $m, n \in \mathbb{N}$. An $\{m \times n \text{ matrix } A\}$ is a “table” of real numbers with m rows and n columns:

$$A = (a_{i,j})_{i=1, \dots, m}^{j=1, \dots, n} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

Addition: For $A = (a_{i,j})_{i=1,\dots,m}^{j=1,\dots,n}$ and $B = (b_{i,j})_{i=1,\dots,m}^{j=1,\dots,n}$ we define

$$A + B = (c_{i,j})_{i=1,\dots,m}^{j=1,\dots,n} \quad \text{where} \quad c_{i,j} = a_{i,j} + b_{i,j} .$$

Multiplication by a scalar: For $\lambda \in \mathbb{R}$ and $A = (a_{i,j})_{i=1,\dots,m}^{j=1,\dots,n}$ we define

$$\lambda A = (d_{i,j})_{i=1,\dots,m}^{j=1,\dots,n} \quad \text{where} \quad d_{i,j} = \lambda a_{i,j} .$$

Zero vector: Define $\underline{0}$ to be the “all zero” matrix.

20 Matrices (continued 1)

Examples for 3×4 -matrices ($m = 3$ and $n = 4$):

$$\begin{pmatrix} 2 & -2 & 0 & 1 \\ 1 & -1 & 1 & 2 \\ -5 & 3 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 0 & 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 3 & 5 \\ 6 & 5 & 8 & 10 \\ -5 & 5 & 2 & 3 \end{pmatrix}$$

$$\lambda \cdot \begin{pmatrix} 2 & -2 & 0 & 1 \\ 1 & -1 & 1 & 2 \\ -5 & 3 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2\lambda & -2\lambda & 0 & \lambda \\ \lambda & -\lambda & \lambda & 2\lambda \\ -5\lambda & 3\lambda & \lambda & 0 \end{pmatrix}$$

$$\underline{0} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

21 Matrices (continued 2)

$\mathcal{M}_{m,n}$ = set of all $m \times n$ -matrices with real entries

With respect to the above structure, $\mathcal{M}_{m,n}$ is a vector space over $\mathbb{R}$.

Proof. Up to re-arrangement, an $(m \times n)$ -matrix is nothing but an mn -tuple of real numbers, and the vector space operations in $\mathcal{M}_{m,n}$ and $\mathbb{R}^{mn}$ correspond under this identification. So $\mathcal{M}_{m,n}$ is a vector space since $\mathbb{R}^{mn}$ is. ■

22 The set of polynomials of degree $\leq n$

For $n \geq 0$ let $\mathcal{P}_n$ be the set of all polynomials with real coefficients of degree at most n :

$$\mathcal{P}_n = \{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \mid a_n, a_{n-1}, \dots, a_0 \in \mathbb{R}\}$$

We define addition and multiplication by a number in the usual way: for $\lambda \in \mathbb{R}$ and

$$\begin{aligned} f &= a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 , \\ g &= b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0 , \end{aligned}$$

we define

$$\begin{aligned} f + g &= (a_n + b_n)x^n + (a_{n-1} + b_{n-1})x^{n-1} + \\ &\quad \dots + (a_1 + b_1)x + (a_0 + b_0) \quad \text{and} \\ \lambda f &= (\lambda a_n)x^n + (\lambda a_{n-1})x^{n-1} + \dots + (\lambda a_1)x + \lambda a_0 . \end{aligned}$$

23 The set of polynomials of degree $\leq n$ (continued)

We denote the zero polynomial by $\underline{0}$, that is,

$$\underline{0} = 0x^n + 0x^{n-1} + \dots + 0x + 0 .$$

With respect to this structure, the set $\mathcal{P}_n$ is an $\mathbb{R}$ -vector space.

Proof. A polynomial of degree $\leq n$ is determined by an $(n+1)$ -tuple of real numbers, *viz.*, its coefficients. Addition of polynomials corresponds to componentwise addition of entries in this tuple, and multiplication by a scalar scales each coefficient with λ . The zero polynomial corresponds to the all-zero tuple. In other words, $\mathcal{P}_n$ is a vector space since $\mathbb{R}^{n+1}$ is a vector space. ■

24 The set of polynomials (all degrees)

Define $\mathcal{P}$ = the set of all polynomials of arbitrary (but finite!) degree. That is, $\mathcal{P}$ is the union of all the sets $\mathcal{P}_n$:

$$\mathcal{P} = \bigcup_{n=0}^{\infty} \mathcal{P}_n = \mathcal{P}_0 \cup \mathcal{P}_1 \cup \mathcal{P}_2 \cdots$$

Equipped with the structure described above (addition of polynomials, multiplication with a scalar, and zero vector), $\mathcal{P}$ is a vector space over $\mathbb{R}$.

This follows from the fact that any two elements of $\mathcal{P}$ lie in the same subset $\mathcal{P}_n$, for large enough n , and that $\mathcal{P}_n$ is known to be a vector space already.

25 Functions on an interval

For $I \subseteq \mathbb{R}$ an interval define

$$\mathcal{F}(I) = \text{set of all functions } f: I \rightarrow \mathbb{R} .$$

For $f, g \in \mathcal{F}(I)$ and $\lambda \in \mathbb{R}$ we define

$$\begin{aligned} f + g: I &\rightarrow \mathbb{R} , & (f + g)(t) &= f(t) + g(t) \\ \lambda f: I &\rightarrow \mathbb{R} , & (\lambda f)(t) &= \lambda \cdot f(t) \end{aligned}$$

Let $\underline{0}$ denote the zero function: $\underline{0}(t) = 0$ for all $t \in I$.

Equipped with this structure, $\mathcal{F}(I)$ is a vector space over $\mathbb{R}$.

26 An unusual vector space

Consider the set of pairs of real numbers, with first entry positive:

$$\mathcal{W} = \{(a, b) \in \mathbb{R} \times \mathbb{R} \mid a > 0\}$$

We *define* addition and multiplication by a scalar as

$$(a, b) + (c, d) = (ac, b + d) , \quad \lambda \cdot (a, b) = (a^\lambda, \lambda b) .$$

We also define $\underline{0} = (1, 0) \in \mathcal{W}$.

With these definitions, $\mathcal{W}$ is actually an $\mathbb{R}$ -vector space!

This example shows that “addition of vectors” need not look like addition at all. What matters is that the properties (A1)–(S4) are satisfied.

27 An even more unusual vector space

Consider the following set of pairs of numbers:

$$\mathcal{E} = \{(a, b) \mid a > 0 \text{ and } b \in \mathbb{R}\}$$

We *define* addition and multiplication by a scalar as

$$(a, b) + (c, d) = (ac, ad + bc) , \quad \lambda(a, b) = (a^\lambda, \lambda a^{\lambda-1}b) .$$

We also define $\underline{0} = (1, 0) \in \mathcal{E}$.

With these definitions, $\mathcal{E}$ is actually an $\mathbb{R}$ -vector space!

This example shows again that vectors spaces may look strange. However, since the properties (A1)–(S4) are satisfied they all “behave in the same way”!

Section 2.3: General properties of vector spaces

28 The vector “ $-v$ ”

V an $\mathbb{R}$ -vector space, $v \in V$ given.

By property (A4) there is $w \in V$ with $v + w = \underline{0}$.

The vector w is uniquely determined by v : If $w_2 \in V$ satisfies $v + w_2 = \underline{o}$ as well, then $w = w_2$.

Proof:

$$\begin{aligned} w &\stackrel{(A3)}{=} w + \underline{o} = w + (v + w_2) \stackrel{(A2)}{=} (w + v) + w_2 \\ &\stackrel{(A1)}{=} (v + w) + w_2 = \underline{o} + w_2 \stackrel{(A1)}{=} w_2 + \underline{o} \stackrel{(A3)}{=} w_2 \quad \blacksquare \end{aligned}$$

The unique vector w is denoted by $-v$.

One can show that $-v = (-1)v$ holds for every vector v of an arbitrary vector space V .

We write $u - v$ instead of $u + (-v)$ so that $u - u = \underline{o}$ as expected.

29 Vector spaces: first properties

Let V be an $\mathbb{R}$ -vector space with zero vector $\underline{o}$.

- (i) For all $u, v, w \in V$, if $u + v = u + w$ then $v = w$.
- (ii) For all $u \in V$ we have $0u = \underline{o}$.
- (iii) For all $u \in V$ we have $(-1)u = -u$.
- (iv) For all $\lambda \in \mathbb{R}$ we have $\lambda \underline{o} = \underline{o}$.
- (v) Let $\lambda \in \mathbb{R}$ and $u \in V$. We have $\lambda u = \underline{o}$ if and only if $\lambda = 0$ or $u = \underline{o}$.
- (vi) Let $u, v \in V$ and $0 \neq \lambda \in \mathbb{R}$. If $\lambda u = \lambda v$ then $u = v$.
- (vii) For all $\lambda \in \mathbb{R}$ and $v \in V$ we have $-(\lambda v) = (-\lambda)v$.

30 Vector spaces: first properties (continued)

Claim: (i) For all $u, v, w \in V$, if $u + v = u + w$ then $v = w$.

Proof.

$$\begin{aligned}
 u + v = u + w & \Rightarrow -u + (u + v) = -u + (u + w) \\
 & \xRightarrow{(A2)} (-u + u) + v = (-u + u) + w \\
 & \xRightarrow{(A1)} (u + (-u)) + v = (u + (-u)) + w \\
 & \xRightarrow{(*)} \underline{0} + v = \underline{0} + w \\
 & \xRightarrow{(A1)} v + \underline{0} = w + \underline{0} \\
 & \xRightarrow{(A2)} v = w
 \end{aligned}$$

At $(*)$ we used the property $u + (-u) = u - u = \underline{0}$ (slide 28 of this chapter). ■

31 Vector spaces: first properties (continued)

Claim: (ii) For all $u \in V$ we have $0u = \underline{0}$.

Proof. We have

$$0u + \underline{0} \underset{(A3)}{=} 0u = (0 + 0)u \underset{(S2)}{=} 0u + 0u .$$

Now using (i), applied to $v = \underline{0}$ and $w = 0u$, we conclude that indeed $\underline{0} = 0u$. ■

32 Vector spaces: first properties (continued)

Claim: (iii) For all $u \in V$ we have $(-1)u = -u$.

Proof.

$$\begin{aligned}
 u + (-u) &= \underline{0} && \text{property of } -u \\
 &= 0u && \text{by (ii)} \\
 &= (1 + (-1))u && \text{since } 1 + (-1) = 0 \text{ in } \mathbb{R} \\
 &= 1u + (-1)u && \text{by (S2)} \\
 &= u + (-1)u && \text{by (S4)}
 \end{aligned}$$

Now part (i), applied to $v = -u$ and $w = (-1)u$, implies that $-u = (-1)u$. ■

33 Vector spaces: first properties (continued)

Claim: (iv) For all $\lambda \in \mathbb{R}$ we have $\lambda \underline{0} = \underline{0}$.

Proof.

$$\begin{aligned}
 \underline{0} &= \lambda \underline{0} + (-\lambda \underline{0}) && \text{by (A4) for } \lambda \underline{0} \\
 &= \lambda(\underline{0} + \underline{0}) + (-\lambda \underline{0}) && \text{by (A3) for } \underline{0} \\
 &= (\lambda \underline{0} + \lambda \underline{0}) + (-\lambda \underline{0}) && \text{by (S1) (distributivity)} \\
 &= \lambda \underline{0} + (\lambda \underline{0} + (-\lambda \underline{0})) && \text{by (A2) (associativity)} \\
 &= \lambda \underline{0} + \underline{0} && \text{since } \lambda \underline{0} + (-\lambda \underline{0}) = \underline{0} \\
 &= \lambda \underline{0} && \text{by (A3) for } \underline{0}
 \end{aligned}$$
■

34 Summary

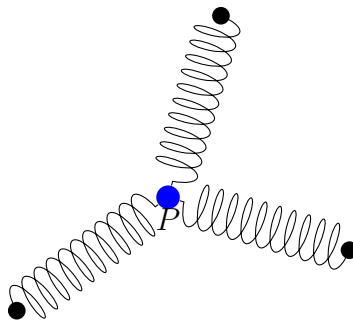
- Vectors (= elements of a vector space) can be added and subtracted.
- Vectors can be multiplied by scalars (= real numbers).
- Vectors “behave” like geometric objects (arrows); one can visualise addition as “concatenation of arrows”, and multiplication by a scalar as “stretching/scaling an arrow”.
- There are many examples of vector spaces, including

- $\mathbb{R}^n$, the vector space of n -tuples of real numbers,
- the vector space of solutions of a homogeneous system of linear equations,
- $\mathcal{M}_{m,n}$, the vector space of m -by- n matrices,
- $\mathcal{P}_n$, the vector space of polynomials of degree $\leq n$,
- $\mathcal{S}$, the vector space of sequences of real numbers,
- $\mathcal{F}(I)$, the vector space of functions $I \rightarrow \mathbb{R}$.
- The abstract notion of a vector space allows us to deal with many examples at the same time; statements need to be proved once only, instead of separately for each special case.

Section 2.4: A sample application of vector spaces in physics

35 Application: equilibrium of forces

A particle P can move on a (horizontal, plane) surface. It is currently at rest in position $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. We use three springs to apply forces to P , as in the following picture:

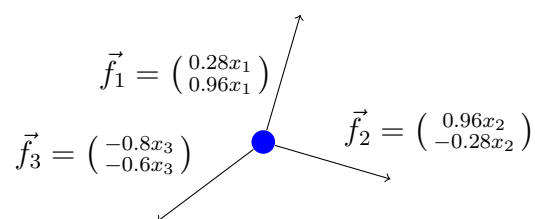


The forces can be represented by elements of $\mathbb{R}^2$ (the motion is restricted to a plane: we are in a “2-dimensional situation”).

Fact: The particle P will remain at rest if the forces cancel out, that is, if the sum of the corresponding vectors is the zero vector.

36 Application: equilibrium of forces (continued)

The forces in our case are represented by vectors $\vec{f}_j$ (using physics notation) with magnitudes x_j (measured in “Newton”):



$$\vec{f}_1 = \begin{pmatrix} 0.28x_1 \\ 0.96x_1 \end{pmatrix} \quad \vec{f}_2 = \begin{pmatrix} 0.96x_2 \\ -0.28x_2 \end{pmatrix} \quad \vec{f}_3 = \begin{pmatrix} -0.8x_3 \\ -0.6x_3 \end{pmatrix}$$

Equilibrium condition: $\vec{f}_1 + \vec{f}_2 + \vec{f}_3 = \underline{0}$

This translates into the following system of linear equations:

$$\left. \begin{array}{rrcr} 0.28x_1 & + & 0.96x_2 & - & 0.8x_3 & = & 0 \\ 0.96x_1 & - & 0.28x_2 & - & 0.6x_3 & = & 0 \end{array} \right\}$$

37 Application: equilibrium of forces (continued)

A single elimination step leads to the following system:

$$\left. \begin{array}{rrcr} 0.28x_1 & + & 0.96x_2 & - & 0.8x_3 & = & 0 \\ & - & \frac{25}{7}x_2 & + & \frac{15}{7}x_3 & = & 0 \end{array} \right\}$$

One particular solution is $x_3 = 1N$, $x_2 = 0.6N$ and $x_1 = 0.8N$.

The system has infinitely many solutions. This is physically meaningful. For example, when the forces cancel out the same will be true on doubling all forces.